

BJT NONLINEAR MODEL PARAMETERS ⁽¹⁾

Parameters	Q1, Q2	Parameters	Q1, Q2
IS	2.7e-16	MJC	0.56
BF	185	XCJC	0.2
NF	1.02	CJS	0
VAF	15	VJS	0.75
IKF	0.055	MJS	0
ISE	1.77e-11	FC	0.5
NE	2.1	TF	5e-12
BR	1	XTF	2
NR	1	VTF	25
VAR	Infinity	ITF	0.001
IKR	Infinity	PTF	0
ISC	0	TR	0.3e-9
NC	2	EG	1.11
RE	0.6	XTB	0
RB	12	XTI	3
RBM	3.7	KF	0
IRB	1.2e-5	AF	1
RC	8		
CJE	1.2e-12		
VJE	0.77		
MJE	0.5		
CJC	0.8e-12		
VJC	0.27		

(1) Gummel-Poon Model

Note:

This nonlinear model utilized the latest data available.
See our Design Parameter Library at www.cel.com for this data.

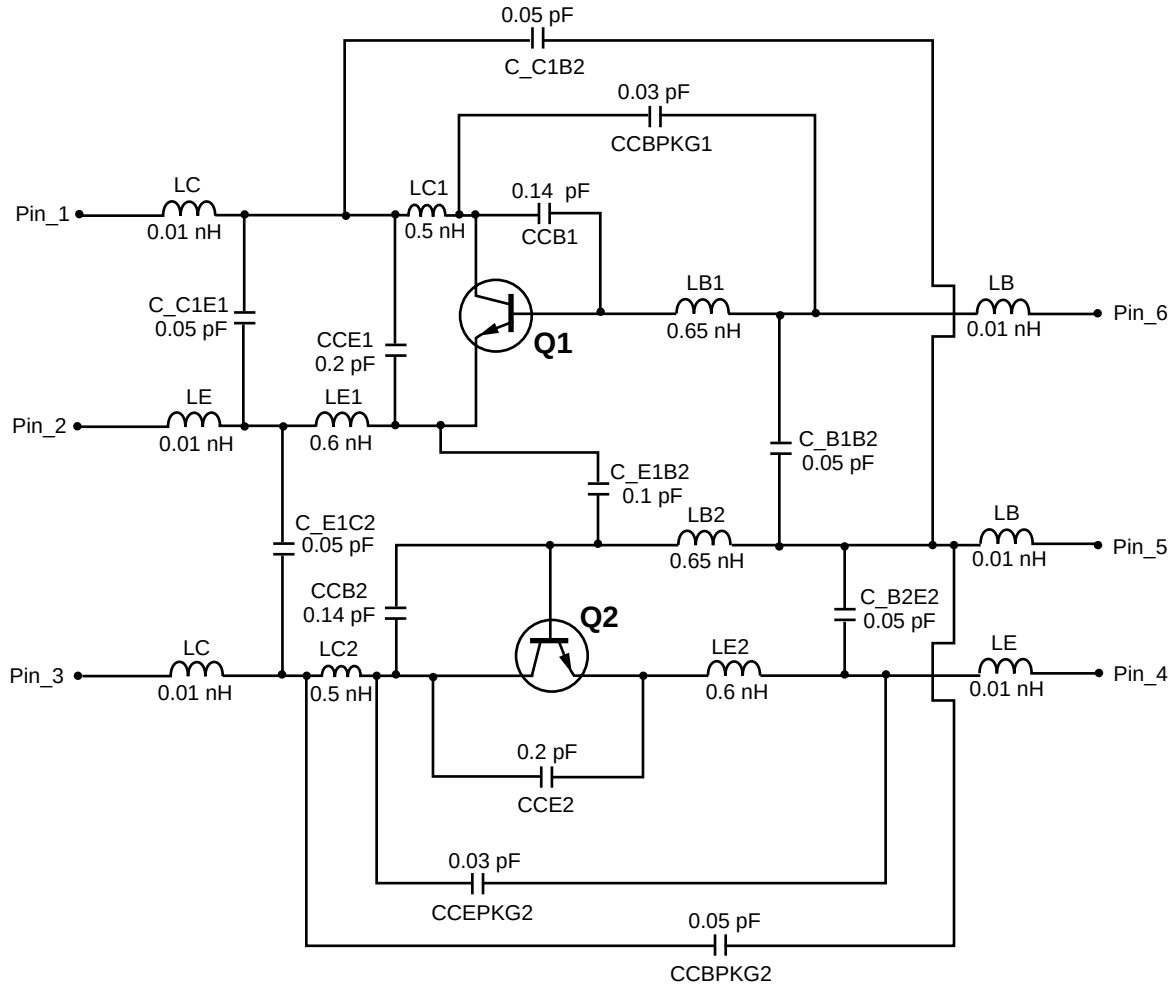
UNITS

Parameter	Units
time	seconds
capacitance	farads
inductance	henries
resistance	ohms
voltage	volts
current	amps

MODEL RANGE

Frequency: 0.1 to 7.0 GHz
Bias: $V_{CE} = 2.5 \text{ V}$ to 8 V , $I_C = 1 \text{ mA}$ to 10 mA
Date: 02/01

SCHEMATIC



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Frequency: 0.1 to 7.0 GHz

Bias: $V_{CE} = 2.5 \text{ V to } 8 \text{ V}$, $I_c = 1 \text{ mA to } 10 \text{ mA}$

Date: 02/01